

Humans’ Fast and Slow Responses to Competition from Internal Generative AI in Online Q&A Communities

Abstract

The advent of generative artificial intelligence (GenAI) has disrupted online question-and-answer (Q&A) communities and prompted debates on how platforms should respond. Our work examines a platform strategy of introducing platform-provided (internal) GenAI answers, an arrangement that differs fundamentally from the external general-purpose GenAI systems (like ChatGPT) studied in prior work: rather than operating privately and beyond a platform’s authority, an internal answerer is deployed under the platform’s own governance, displayed under an explicit machine-authorship label, and positioned alongside human contributors as an identified competitor. Taking a human-AI competition view, grounded in the two systems of thinking (fast-and-slow) theory, we argue that the total AI effect is the sum of two separable components: a fast response to the AI label and a slower response to the AI content. Using a randomized field experiment at a leading technical Q&A community, we show that the AI label invites human contribution because the community is skeptical of AI’s capabilities, whereas longer AI content suppresses human contribution, and higher-quality AI content raises the quality of human answers as contributors reuse and build on it. Decomposing the total AI effect offers a theory-driven framework for reconciling seemingly conflicting findings in the literature and for anticipating human responses as AI capabilities advance. We contribute to the emerging literature on GenAI’s impact on human society and provide practical insight for the adoption and application of GenAI on platforms.

Keywords: Generative AI, Human-AI Interactions, Online Q&A Communities, Platform Governance, Two Systems of Thinking.